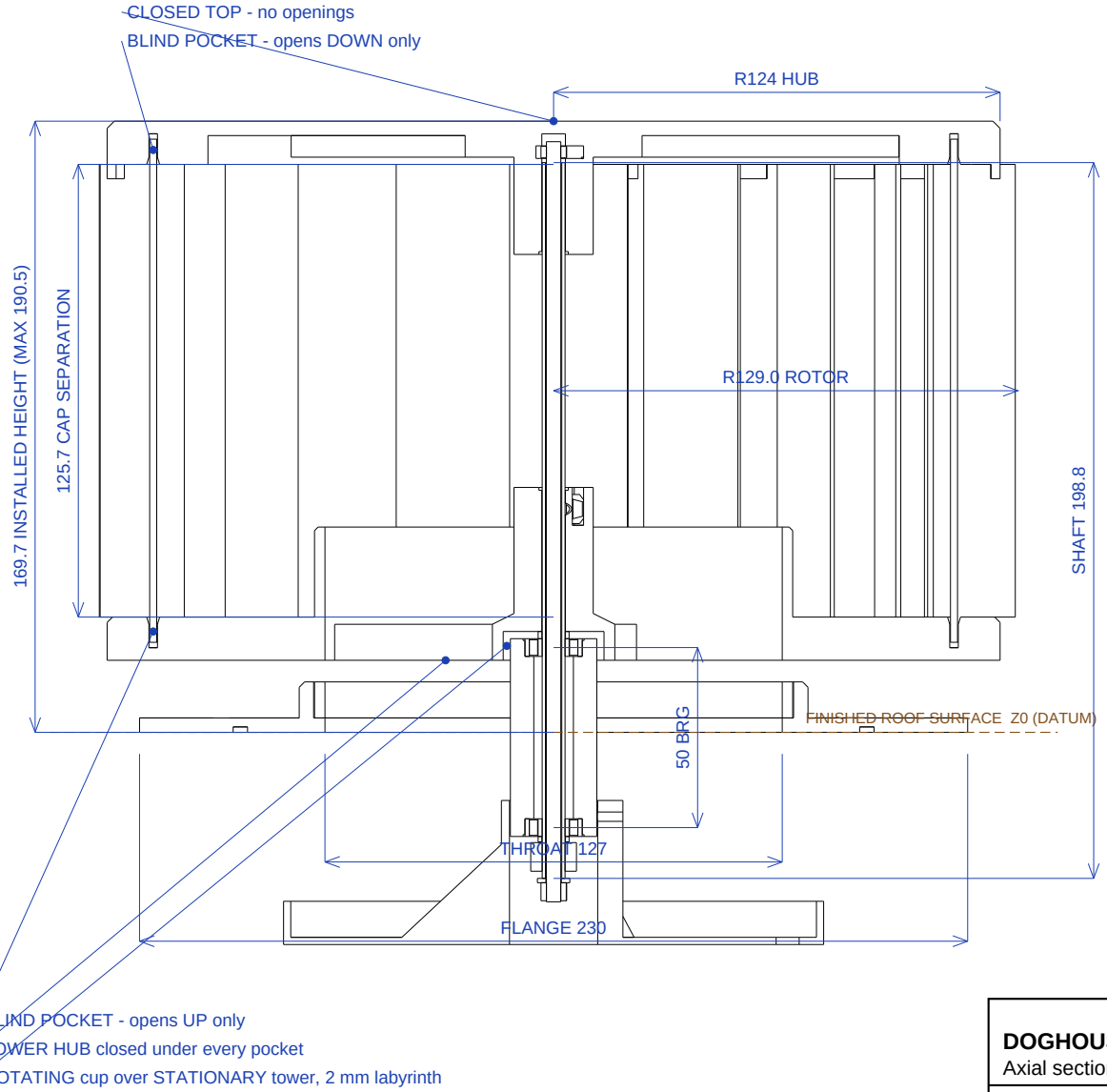
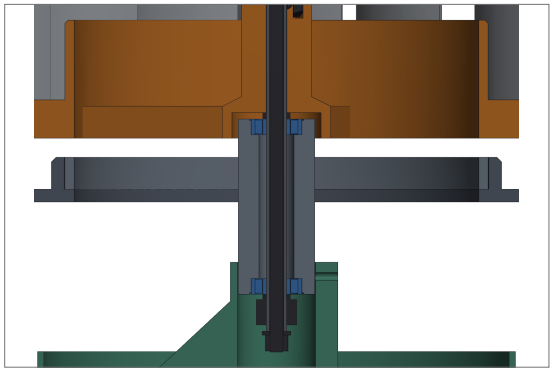
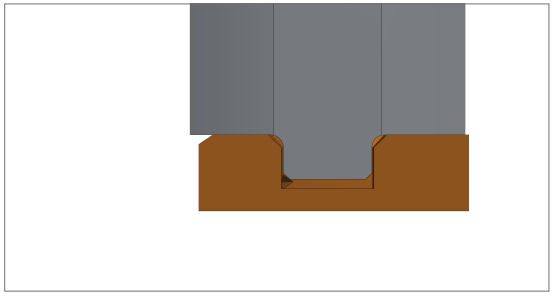
SHEET 2 - AXIAL HALF-SECTION (plane XZ, true CAD hidden-line projection, 1:2) + HEIGHT STACK



Bearing / labyrinth detail (render)



Lower pocket detail: tab, R2 roots, blind floor, weep



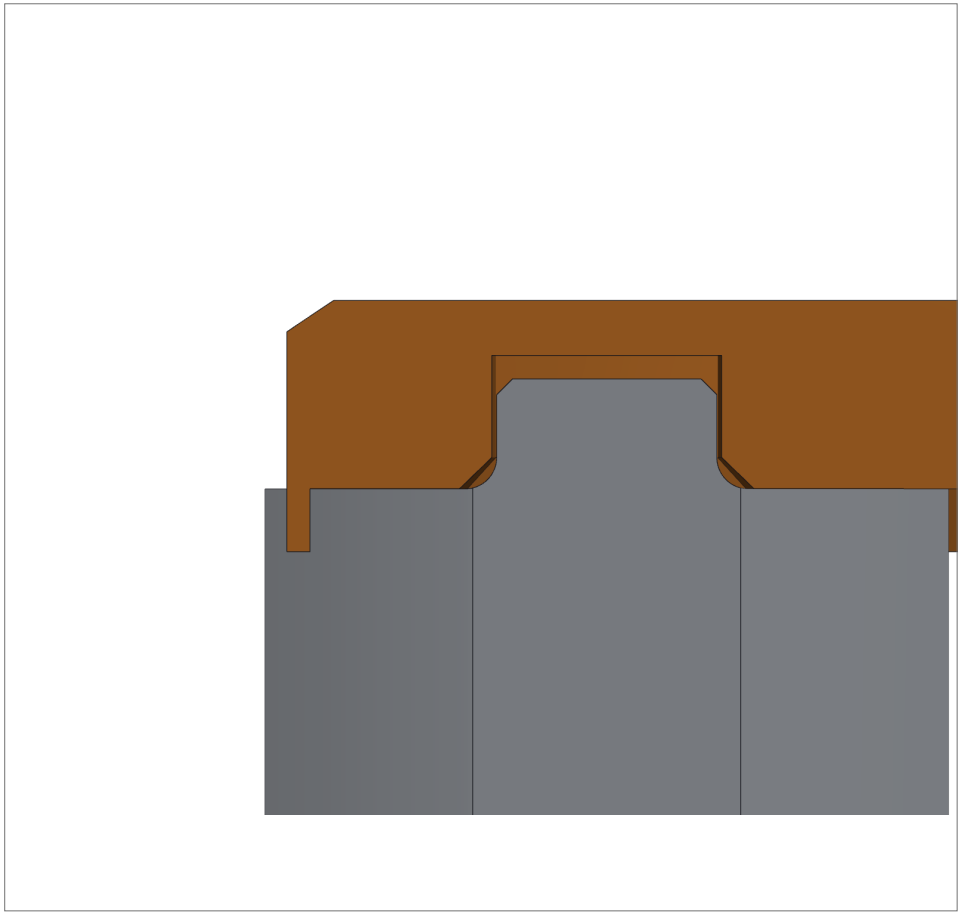
Height stack (Z, mm)	from	to	size
Roof to collar/support top	0	14.0	14.0
Rotor-to-stationary gap	14.0	20.0	6.0
Lower hub	20.0	32.0	12.0
Blade shoulders (cap sep.)	32.0	157.70	125.70
Upper hub	157.70	169.70	12.0
Bearing centres	-26.46	23.54	50.0
Shaft	-40.56	158.20	198.76

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Axial section

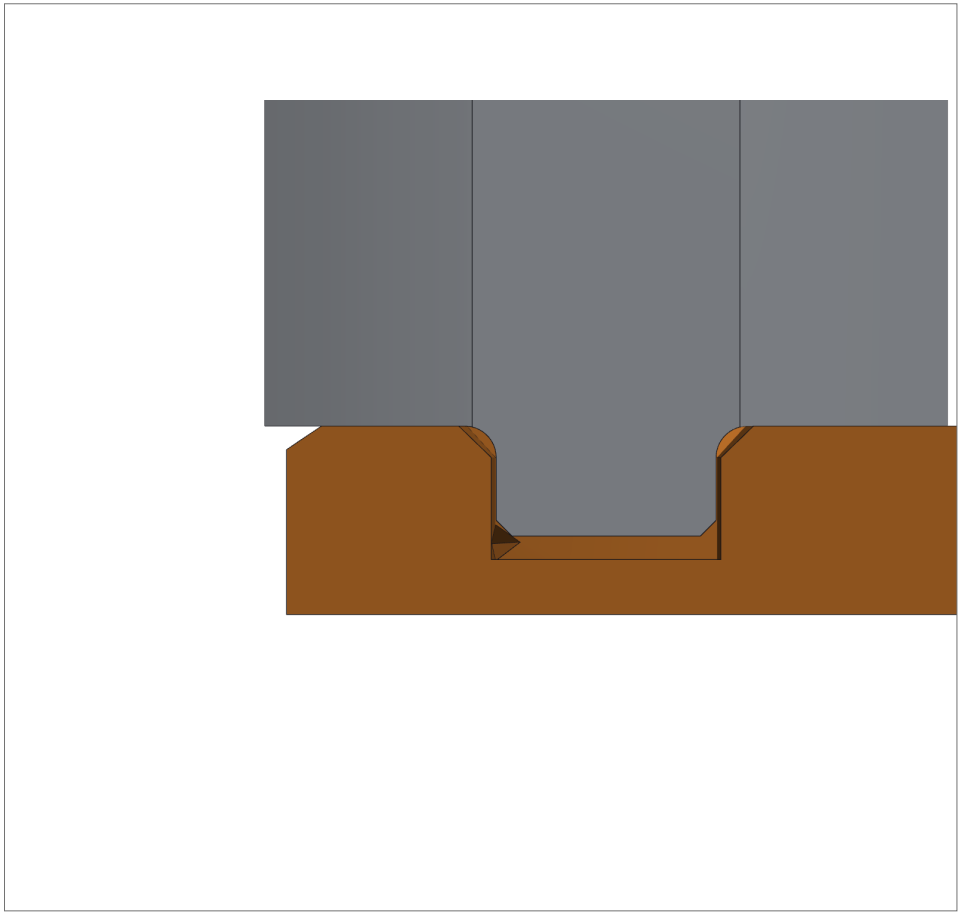
REV DW-V2.0 UNITS mm SCALE 1:2 2026-10-24

SHEET 1/2 Source: build123d parametric model + MASTER_PARAMETERS.json

BLIND POCKET SECTIONS - plane along blade-1 chord through the tab centre (true CAD)



UPPER HUB: pocket opens DOWN only; 3.5 mm intact roof; closed toward OD



LOWER HUB: pocket opens UP only; 3.5 mm floor; diamond weep to OD

Tab 1.60 in pocket 1.95: 0.175 mm per side. Tab 14.0 in pocket 14.5: 0.25 mm per end. Tip to floor 1.5 mm.
Mouth: 2 x 45 deg end lead-in clears the R2 tab-root fillets; 0.5 mm side lead-in guides the tab.
Retention: shoulders seat on the hub faces (axial); closed pocket ends/sides (radial + tangential); hubs fixed to the shaft by set screws in drilled dimples (positive) - so blades cannot escape without removing a hub.

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Blind pocket sections

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

SHEET 2/2 Source: build123d parametric model + MASTER_PARAMETERS.json